

WICHITA JABARA, KS, USA

WMO: 724504

Lat: 37.746N

Lon: 97.221W

Elev: 433

StdP: 96.23

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 03974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.4	-19.4	0.7	-11.9	-17.2	0.9	-9.9	14.2	3.5	12.7	4.0	4.6	350	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	37.5	23.3	35.9	23.4	33.8	23.3	25.4	32.5	24.9	32.0	24.2	31.4	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.3	28.9	22.8	18.4	28.2	22.3	17.9	27.6	80.4	32.6	77.8	32.0	75.3	31.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.4	11.2	10.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-17.5	39.8	3.2	1.9	-19.8	41.1	-21.7	42.2	-23.5	43.3	-25.9	44.6		
			-17.8	26.8	3.2	0.8	-20.1	27.4	-21.9	27.8	-23.7	28.3	-26.1	28.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	0.7	2.9	8.2	13.7	19.1	24.2	27.0	26.0	21.6	14.9	7.9	1.7	(6)
(7)		DBStd	10.54	6.33	6.52	6.12	5.23	4.40	3.34	3.31	3.40	4.16	5.23	5.59	5.66	(7)
(8)		HDD10.0	1012	292	211	106	24	1	0	0	0	17	103	260		(8)
(9)		HDD18.3	2474	547	432	317	154	47	2	0	0	16	131	314	514	(9)
(10)		CDD10.0	2491	3	11	49	134	282	427	527	497	349	167	39	4	(10)
(11)		CDD18.3	912	0	0	3	14	69	179	269	239	114	23	1	0	(11)
(12)		CDH23.3	9840	0	1	26	151	610	1837	3193	2671	1124	218	9	0	(12)
(13)		CDH26.7	4551	0	0	3	33	193	787	1677	1344	455	58	1	0	(13)
(14)	Wind	WSAvg	4.9	5.0	5.1	5.6	5.9	5.0	4.7	4.3	4.0	4.3	4.7	5.0	4.9	(14)
(15)	Precipitation	PrecAvg	897	19	34	60	81	133	145	105	102	78	70	42	27	(15)
(16)		PrecMax	1292	67	105	119	188	281	276	231	232	262	207	137	86	(16)
(17)		PrecMin	631	2	1	11	14	41	52	13	10	14	0	2		(17)
(18)		PrecStd	178	16	25	34	43	64	62	55	57	62	44	40	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	19.1	22.5	26.3	30.0	33.1	36.7	40.1	39.8	36.4	31.2	24.2	18.7		(19)
(20)		MCWB	10.6	12.5	15.6	18.2	21.0	23.1	23.1	23.1	22.1	19.9	15.0	12.8		(20)
(21)		DB	16.0	18.9	23.1	27.3	30.9	34.5	37.8	37.4	33.0	28.0	22.2	15.9		(21)
(22)		MCWB	8.7	11.0	15.1	17.3	21.4	23.5	23.5	23.1	22.2	18.7	14.6	9.8		(22)
(23)		DB	13.0	16.1	20.8	24.9	28.8	32.7	36.2	35.5	31.4	25.9	19.1	12.8		(23)
(24)		MCWB	7.2	9.7	13.7	16.4	20.5	23.2	23.7	23.4	22.0	17.5	12.9	8.3		(24)
(25)		DB	10.3	12.9	18.1	22.6	27.0	31.2	33.9	33.0	29.1	23.2	16.9	10.4		(25)
(26)		MCWB	5.7	7.8	12.2	15.4	20.0	22.9	23.8	23.3	20.8	17.1	11.9	6.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	13.0	15.0	18.8	20.4	23.8	26.1	26.4	26.2	24.7	22.6	17.8	15.1		(27)
(28)		MCDWB	15.5	18.0	23.8	26.7	29.4	32.7	33.5	32.6	31.3	27.4	21.8	17.9		(28)
(29)		WB	10.0	12.7	16.9	19.4	22.8	25.1	25.5	25.2	23.6	21.2	15.6	11.2		(29)
(30)		MCDWB	14.8	16.9	20.5	24.4	28.4	31.7	32.8	32.5	30.2	25.3	19.5	13.5		(30)
(31)		WB	7.7	10.5	15.2	18.0	21.8	24.3	24.9	24.6	22.8	19.6	14.3	8.7		(31)
(32)		MCDWB	12.3	16.1	19.2	22.8	26.9	30.7	32.3	31.7	29.2	23.4	18.0	12.3		(32)
(33)		WB	5.8	7.9	12.8	16.6	20.8	23.5	24.3	23.9	22.0	17.9	12.5	6.6		(33)
(34)		MCDWB	10.3	12.6	17.7	21.7	25.6	29.4	31.9	30.8	27.7	21.9	16.4	10.3		(34)
(35)	Mean Daily Temperature Range	MDBR	11.2	11.6	12.3	12.2	11.4	11.2	11.8	11.8	12.1	12.2	11.7	10.4		(35)
(36)		5% DB	15.4	16.0	15.7	15.0	13.5	12.5	14.0	14.3	13.6	14.8	14.6	14.2		(36)
(37)		MCWBR	9.5	9.9	9.0	8.3	6.3	4.8	4.0	4.2	5.3	7.1	8.4	9.4		(37)
(38)		5% WB	14.6	14.8	12.7	13.0	11.6	11.4	11.9	12.0	11.7	11.4	12.0	13.0		(38)
(39)		MCWBR	9.7	9.9	8.6	8.3	6.4	5.3	4.6	4.7	5.1	6.9	8.3	9.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.298	0.332	0.373	0.407	0.412	0.420	0.424	0.379	0.336	0.297	0.280		(40)
(41)		taud	2.501	2.475	2.411	2.319	2.297	2.312	2.331	2.276	2.377	2.486	2.526	2.525		(41)
(42)		Ebn at Noon	909	939	932	906	877	869	858	847	869	882	887	892		(42)
(43)		Edh at Noon	78	92	108	126	131	129	126	130	111	90	76	72		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.50	3.23	4.22	5.31	5.96	6.80	6.70	6.00	4.99	3.69	2.72	2.15		(44)
(45)		RadStd	0.25	0.46	0.29	0.35	0.52	0.49	0.41	0.34	0.36	0.42	0.27	0.20		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.37	+0.47	N/A	N/A	N/A	N/A	(46)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page